

5 Interesting Reference Designs FOR SENSORS

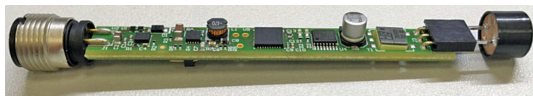
Designing sensors is tough. Ready-made reference designs for industrial, automotive, and health applications help you prototype faster.

NIDHI AGARWAL

When designing industrial systems, challenges often arise in fitting sensors into compact spaces, managing power consumption, ensuring reliable communication, and maintaining accuracy in demanding environments. Reference designs address these issues through tested hardware and software that save time and minimise errors. This selection highlights reference designs tailored for specific sensing and control tasks, from ultrasonic distance measurement and multi-sensor networking to current, health, and sensor interfacing. Each helps accelerate prototyping, streamline validation, and enhance overall system design.

Ultrasonic distance sensor

Have you ever tried fitting ultrasonic sensing into a tiny space like an M12 housing and struggled with dust, temperature swings, electrical noise, or vibrations affecting your measurements? We all want accurate readings, low power use, real-time communication, and protection from power faults without making the design more complex or expensive. That is



why you can use a reference design from Texas Instruments. It is compact, fits inside an M12 housing, and saves us time while reducing errors. Inside, it combines a piezoelectric transducer for sensing, a centre-tap transformer for voltage boost, and a system-on-chip handling the analogue front-end, programmable-gain amplifier, ADC, and digital signal processing, while a microcontroller manages timing, data processing, and protocol tasks, and a transient-voltage suppressor protects against surges, giving us a ready-to-adapt blueprint for reliable, efficient ultrasonic sensing in tough industrial conditions.

Applications. You can use this design to develop sensors suitable for multiple industrial scenarios, such as robots in logistics warehouses, programmable industrial robots, displacement and level measurement systems, or drones.

Link. <https://www.electronicsforu.com/electronics-projects/industry-grade-projects/ultrasonic-distance-sensor-reference-design>

OEM Brand. Texas Instruments (TI)

Industrial multi-sensor

When we work on industrial sensor and actuator networks, we often struggle with driving loads or data lines efficiently, keeping power use low, maintaining protection, and fitting everything into a design.



The Industrial Smart Sensor reference design from STMicroelectronics helps us tackle these challenges by providing a hardware and software solution for IO-Link industrial applications. The microcontroller at the heart of the system runs IO-Link control software, manages communication between the dual-channel transceiver and MEMS sensors via standard interfaces, and integrates IoT functions. The transceiver provides reverse polarity protection and EMC immunity, so we don't have to spend extra time designing for industrial conditions. The board integrates a 3-axis magnetic sensor and an inertial module with a machine learning core, both providing digital output.